

Nick Young

Email: youngnicholas228@gmail.com | Website: n-young.me | Github: [n-young](https://github.com/n-young) | LinkedIn: [in/ndashyoung](https://in.linkedin.com/in/ndashyoung)

Education

Brown University [GPA: 3.94/4.00] - **Concurrent MS/BS in Math and Computer Science** May 2023

- > Head TA for revamped [Databases Management Systems course](#). Designed assignments in GoLang for SQL, memory, hash indexes, B+Tree indexes, query optimization, concurrency, and recovery.
- > Head TA for new [Applied Cryptography course](#). Designed assignments in C++ for: communication, authentication, zero-knowledge, multi-party computation, and homomorphic encryption.
- > Published paper on [Adversarial Thinking in CS Education](#) in ACM ICER 2021.
- > Relevant Coursework: Multiparty Computation, Cryptography, Number Theory, Graph Theory, Algebra, Massive Time Series Databases, Database Management Systems, Distributed Systems, Networking, Prescriptive Analytics, Formal Methods, Programming Languages, Compilers, Graphics.

Experience

Databricks - Software Engineer (Engine)

September 2025 - Present

Promoted to SWE II in early-2025; nominated to newly-formed engine rewrite team in mid-2025.

- > Expanded analyzer coverage for aggregates, windows, grouping sets, and set operations.
- > Designed various logical planning infrastructure, including plan hashing, plan equality, deep cloning, planning metrics, variadic expressions, plan indexing, and plan serialization.
- > Imported and instrumented millions of queries from open source for correctness testing.
- > Implemented custom linters enforcing correctness and code cleanliness across the team.

Databricks - Software Engineer (Query Optimization)

August 2023 - Present

- > Designed and shipped dynamic file/partition pruning for shuffle/broadcast joins achieving 20x scan time reduction in benchmarks; DFP/DPP now injected on ~10M queries/day across the fleet.
- > Tuned cost models and statistics models for a variety of features (file pruning, partition pruning, Bloom filters) and operators (windows, filters, joins, limits) with data experiments.
- > Architected and implemented correctness runtime validation of Bloom filters, validating trillions of rows and millions of filters daily; unblocked Blooms for DML statements.
- > Designed and implemented automated plan fallback, with fine-grain control over recovery from rule or batch, automatically mitigating dozens of unique potential SEV 1 availability incidents.
- > Improved optimizer stability across ~6M streaming pipelines, directly unblocking 5+ customers.
- > Devised window and aggregate optimizations achieving up to 99% speedup on ~10k queries/day.
- > Optimized memory by 16 bytes/plan node, alleviating driver pressure across all queries.
- > Served as primary on-call; fixed issues across Spark's optimizer, execution, stats, and core.
- > Led SEV 0 correctness incident response with fix, post-mortem, and reusable tooling.

Stripe - Software Engineering Intern (Databases Fleet Management)

June - August 2022

- > Built distributed MongoDB load testing framework in GoLang supporting 100,000 writes/sec to evaluate fleet capacity and identify performance bottlenecks across database clusters.

Bloomberg LP - Software Engineering Intern (Developer Experience)

June - August 2021

- > Deployed adaptive connection pool in Python reducing new connections by 98% and p50 latency by 50%; instrumented with telemetry and Grafana dashboards for on-call diagnostics.

Skills

Languages: Rust, Scala, GoLang, Python, C++, SQL.

Systems: Apache Spark, Compiler Design, Query Optimization, Query Execution.

Specializations: Data Pruning, Cost-based Optimization, Query Plan Rewrites, Performance Tuning.